

Detim Zhao

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Education

Bachelor of Science, Software Engineering – May 2026

- Arizona State University, Polytechnic, AZ – GPA: 3.94/4.0
 - Barrett (Honors), Dean's List (Fall 2022, Spring 2023, Fall 2023)

Skills

- Programming – Java, Python, C, C++, PostgreSQL, MySQL
- Libraries – pandas, scikit-learn, matplotlib, seaborn, NumPy
- Tools – Git, GitHub, LaTeX, Photoshop, Illustrator, Excel, Gazebo, Robot Operating System, Mavsdk, PX4 Autopilot

Work Experience

Undergraduate Researcher – Texas A&M-Corpus Christi / Corpus Christi, TX 6/5/23-8/11/23

- Collaborated alongside a project partner, mentor, and graduate students to research and innovate a novel human-swarm (drone) interaction.
- Partnered to formulate and develop a Python program to operate multiple PX4 drones in a gazebo simulation through terminal controls. (Python)
- Applied preprocessing to two datasets, then trained and tested five ML models (KNN, SVM, Random Forest, Naive Bayes, Ensemble) for gesture recognition. (Python)
- Co-authored a research paper for publication (DOI: [10.1145/3565287.3617621](https://doi.org/10.1145/3565287.3617621)) that documents findings and work (LaTeX)

Projects

Gesture Recognition of Hand Gestures with ML

- Streamlined data preprocessing, including data cleaning and handling missing values on small and large datasets, enhancing nearly 15% of data for ML usage.
- Trained and evaluated five different ML models on small and large datasets, enhancing model accuracy by about 10% through hyperparameter tuning. (Python, scikit-learn)

Control Drone (PX4) Swarm in Gazebo Simulation

- Designed a script to automate setup for PX4 drones and Gazebo, saving time from manual inputs and optimizing future testing. (Python)
- Implemented terminal input commands to control individual drones or groups of drones.

Console-Based Chat Application

- Developed console-based directions and options for users to register, log in, update info, create a chat room, join a chat room, and log out.
- Implemented commands when the user is in a chat room to display active users, chat history, and more.
- Created and Organized local host pSQL database to store information about the user and messages. (Java, PostgreSQL)

Maze Generator and Solver

- Developed a GUI-based maze generator and solver where the user specifies the dimensions of the maze.
- Implemented a recursive backtracking algorithm to solve the maze and draw the solution path. (Java, JavaFX)